Eric Ming Tang, Ph.D.

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TECHNICAL SKILLS (*Experience*)

Programming Languages (8 years): Python, C/C++ (multithreaded, real-time), MATLAB, Java

Machine Learning & AI (6 years): Deep learning (PyTorch/LibTorch, TensorFlow), CNNs (U-Net, YOLOv4), GANs, object detection and tracking, clustering and dimensionality reduction (Scikit-learn)

Edge Deployment: PyTorch model export via ONNX, TensorRT optimization, real-time GPU-accelerated inference on NVIDIA Jetson Orin Nano

Computer Vision (8 years): Signal and image processing (filtering, Fourier-domain reconstruction, correlation), denoising, super-resolution, image fusion, stereo vision, camera calibration, 3D reconstruction

Software Tools (8 years): Microsoft Visual Studio, VS Code, Qt Creator, LabView, OpenCV, Scikit-learn

Hardware (8 years): Control systems, DAQ/FPGA programming, camera sensor characterization, optical system design and optimization

EDUCATION

Vanderbilt University, *Ph.D. in Biomedical Engineering* — 2022

Duke University, *Bachelor of Science in Biomedical Engineering* — 2018

Duke University, *Bachelor of Arts in Computer Science* — 2018

WORK EXPERIENCE

Topcon Healthcare — San Jose, CA

Senior Research Scientist — May 2023 – Present

- Managed and optimized real-time image acquisition software (C++) for high-throughput data capture at 2 gigasamples/second and flexible customization of imaging workflows via multithreaded programming
- Designed a physics-informed deep learning model (PyTorch) for image super-resolution and denoising, integrating the optical system's PSF and acquisition parameters, achieving a 2x improvement in resolution and signal quality; results were presented to company executives and stakeholders
- Deployed deep learning imaging models to embedded edge hardware by exporting PyTorch models via ONNX and optimizing with TensorRT for real-time inference on an NVIDIA Jetson Orin Nano
- Led cross-organizational collaborations between vendors and engineering teams to co-develop a novel compact imaging sensor and scale it from development to production, achieving a 2x reduction in device size; contributed to hardware integration, software evaluation, and quantitative benchmarking
- Evaluated and enhanced third-party analysis software by identifying algorithmic limitations, technical problem solving, providing performance feedback, and iterating improvements

Diagnostic Imaging and Image-Guided Interventions Laboratory — Nashville, TN

Postdoctoral Research Fellow — May 2022 – May 2023

- Extended and adapted a deep-learning-based pipeline for 4D surgical instrument tracking by implementing multi-channel inputs and data augmentation, improving model robustness and accuracy in diverse imaging conditions and achieving 99% sensitivity at 23 fps
- Investigated advanced machine learning models (MATLAB/Python: multiscale context aggregation networks, pix2pix, cGANs) to improve neural network input quality by up to 4x under challenging surgical conditions
- Supported development, training, and real-time deployment of an image denoising framework utilizing multi-scale U-Net architectures to improve image quality by 90% with video-rate performance at 22 fps

Graduate Research Assistant — May 2018 – May 2022

- Optimized a high-speed multimodal imaging system and managed acquisition software codebase (C++) that supported real-time processing, live preview of reflectance and 3D images, and GPU-accelerated CNN architectures at 4 gigasamples/second
- Integrated an automated deep-learning-based framework for surgical instrument tracking using multi-view inputs via a lightweight model (YOLOv4) for real-time image acquisition and object detection at 120 fps

SELECTED PUBLICATIONS (Google Scholar)

1. **Tang, E. M.,** El-Haddad, M. T., Patel, S. N., and Tao, Y. K., "Automated instrument-tracking for 4D video-rate imaging of ophthalmic surgical maneuvers," Biomedical Optics Express 13(3), 1471–1484 (2022).
2. Rico-Jimenez, J. J., Hu, D., **Tang, E. M.**, Oguz, I., and Tao, Y. K., "Real-Time OCT Image Denoising Using Self-Fusion Neural Network," Biomedical Optics Express 13(3), 1398–1409 (2022).
3. **Tang, E. M.** and Tao, Y. K., "Modeling and optimization of galvanometric point-scanning temporal dynamics," Biomedical Optics Express 12(11), 6701–6716 (2021).